Harsh Dugar

EXTC

Company: Quantiphi

Job Profile: Machine Learning Engineer

Offer Type: Internship + Placement (Performance Based PPO)

Internship Stipend: 24k

CTC: 8.5 LPA - 6.5L (Base) + 2L (Bonus/QCDP)

Location: Mumbai/Bangalore

Process Date: 21st July OA & 22nd July Interviews

Resume: **Harsh_Dugar_Resume.pdf** (As of Aug 2025)

Placement Process:

The process had 4 rounds.

1) OA
2) Technical Interview 1
3) Technical Interview 2
4) HR Interview

Quantiphi offered 3 roles, FE (Framework Engineer), MLE (Machine Learning Engineer) and BA (Business Analyst). We were supposed to order our preference in the Superset Application.

(To read the ML Interview Experience skip over to page 3)

1. **OA** [133 Appeared]

We had 3 sections

1) Aptitude, Logical & Verbal Reasoning: 40 Questions - 30 Minutes
2) CS Fundamentals: 33 Questions - 25 Minutes
3) Coding Problems 3 Questions - 30 Minutes

The amount of time was limited, so speed was the key to getting through in this round.

The first section was mostly as expected, but there were quite a few difficult problems on permutations and a few other topics. There were also a lot of problems on Area and Volume calculations (What is the profit if you sell the material extracted from a pyramid (and a cone) for 10 Rs/cm^3 and you bought the pyramid for 200). But I had to skip most of these because the time was limited and we didn't have calcis allowed. **My advice would be to attempt this section in multiple iterations, solving the easiest questions in the first one, then the next and so on. Can't stress this enough, but time is a huge constraint here so manage it well.**

The second section was CS fundamentals (and little dev too). We had general theory based questions from OOPs, DBMS and OS. This was pretty standard. There were unexpected questions though that were the you know it or you don't type.

The final Section had 3 coding problems. An easy one, an easy-medium and a difficult one.

- A thief wants to rob all the houses in a village. There is an array of the number of houses in each neighbourhood and the maximum amount of time(hours) he can take. At what minimum speed in houses per hour does he need to rob to clean the entire village ?
  You just need to take the sum of the houses and divide this number by the total time to get the rate of robbing. Round it off to the higher number and return the integer answer.
- A question very similar to the House Robber Problem from Leetcode. It was a basic include-exclude DP problem with one if condition. I didn't even need to apply DP as the first Recursive code that I ran passed all testcases. Link
- You are given an undirected graph of n islands, each with a loot value, and edges representing travel paths between islands with their travel times. A pirate may start from any island, traveling along available paths, and can visit each island at most once. The total loot is the sum of the visited islands' values, while the total time is the sum of the travel times along the chosen path. Find the maximum loot the pirate can collect.

I was able to solve the first 2 problems but ran out of time attempting the 3rd one.

2. **FE Interview 1** [17(FE) + 17(MLE) + 58(BA - GD followed by interviews)] [4:30 PM]

I had mistakenly selected FE as my first preference so I informed my FE Interviewer that I wanted MLE. We had been notified of this inward mobility for Roles in the Presentation. I was asked to complete this interview after which my request would go on to be processed.

We started with this round with me introducing myself and the interviewer going through my resume. We discussed all my projects (2 ML, 2 Dev) one after another and he asked plenty of questions as I was explaining. Be thoroughly prepared with your projects and don't fake anything. You should be prepared to answer questions like "Why did you use this service and not the other (Postgres vs Mongo, AWS vs Azure, etc) ?", "Are there any live applications that provide this service, how does your project compare to theirs ?", "What changes would you make if you wanted to serve 10k users with your project (Scalability is a very popular question) and other questions that question the inner mechanics of your project like "What purpose does this feature resolve ?", "Couldn't you have simplified things this way, your approach is very inefficient." and so on. Not every interviewer goes in this deep but there are ones that do.. (Baker □)

After the resume we started with his questions. He didn't ask any really difficult questions on DSA or SQL that needed time, this was more of an itinerary of easy to medium theory and coding problems. Here are a few of those:

- Do you know about normalization ? What is the difference between 1NF and 3NF normalization ?
- What is polymorphism ? How do we implement it ?
- Can you tell me about Indexing ? What are the drawbacks of Indexing ?
- Write an SQL query to remove duplicate entries from a database. (I used row_number() and partitioned by PK, after which I removed all entries with a row_num higher than 1)
- Can you do this problem without using windowing functions ?

There were quite a few questions around this difficulty level and I was able to answer most of them. After this round I was asked to appear for Interview 2 where I was informed that I will now be taken to their ML Engineer to appear for MLE Interview 1.

3. **MLE Interview 1 & 2** [6:30 PM & 7:30 PM]

I would like to start this section with an overview of what I think of ML Interviews. These are just the views of a guy that has given 2 ML interviews so read them with a grain of salt.

There is no defined structure to gauge a candidate's ability for an ML position at the graduate level. So Interviewers are going to ask you a very wide variety of questions and your selection depends on how much you can impress them across the duration of the exchange. Here is my personal order of importance for the subjects of your interview:

1) **Your Projects:** Can't stress this enough, but your projects are your best shot at showcasing your skills to the interviewer and showing how deep of an understanding you have in the concepts you've used in these projects. The interviewer can't just ask you to explain the architecture of a GNN Model to test you because that isn't something you can expect the average candidate to know.

   You need to dominate this discussion and tell the interviewer of every sellable skill you've utilized. Tell him about the nature of the dataset you used, about the challenges you faced in feature engineering, about the tools you used to clean/filter the data and augment samples. **Tell him about why you chose your Model over the alternatives** (for me it was why I chose BiLSTM over LSTM and GRU) and about the architecture of your model and what understandings you derived while working on the model and so on.

2) **Fundamentals:** The interviewer will then start asking questions to test your fundamentals such as
   - What is regression ? How do you differentiate between it and classification ?
   - What do you mean by clustering ?
   - What will you do if your dataset has a lot of missing values ?
   - Why do some models with good training accuracy have bad test accuracy ?
   - What is bias vs variance tradeoff ?
   - What are false positives and negatives ?
   - What are precision and recall and when do they become useful ?
   - What metrics did you use to judge your model and explain the significance of each metric ? (I answered the confusion matrix, classification report containing accuracy, precision, recall, f1-score, and the Learning curve comparing training and validation sets)

   There will be plenty of questions like this and they can be prepared for in advance.

3) **Your Knowledge of ML Models:** In the midst of the questions focusing on fundamentals, the interviewer also kept asking me questions on popular models. These included
   - What is Linear Regression ? Is it related to Logistic regression ?
   - Please explain how a Decision Tree works ? Can you draw an example for us ?
   - Why do we need Unsupervised Learning ? Can you talk about any popular models Unsupervised Learning holds ?
   - What is an ensemble model ?
   - What will you do when faced with an Image based dataset.

   Having a basic understanding of these models is sufficient, you don't need to be an expert in everything.

I think this discussion includes most of the questions asked to me during the interviews. I'll list the few remaining questions here:

- The interviewer asked me if I had deployed any models ?

- He asked me if I had worked with NLP ? (I started talking about the need to process words and the Bag-of-Words Model. I had very surface level knowledge on this matter..)
- He then started to ask more detailed questions related to NLP but I was unable to answer them, from his body language it seemed like he expected a little more. In popular ML culture we see everyone talking about Deep Learning but NLP spans a whole lot of projects too and is not as much talked about. Can read a few things up.
- He presented to me a situation where I had a dataset of sentences that were classified as good and bad and we needed a model that could predict sentences in this manner.

This concludes almost all of what I was asked in MLE Interviews 1 and 2. I haven't separated the interviews into two sections because the flow of the interviews was very similar and it didn't make sense to separate them. The only difference was that the second interview didn't involve as many fundamental questions as the first one and was a little more intense. There was a bit of grilling involved in both the interviews and the first one went on for over an hour. The second one was a bit short at 30 minutes.

This was followed by the HR interview. The selection from FE/MLE/BA Interview 1 to Interview 2 was very brutal. The rate of progression from Interview 1 to Interview 2 at Quantiphi was far lower than the average company.

4. **HR Interview** [13 shortlisted, 3 MLE and 10 FE+BA] [8:30 PM]

This was a typical "HR Interview". There weren't any tough questions, just enquiring things about you. Questions included. It lasted a relaxed 15-20 minutes.

After the HR interview we had to wait for almost an hour as all the Interviewers and Execs discussed amongst themselves. Finally the shortlist was released around 10 PM and 8 candidates were selected. The selectees were 2 from MLE, 3 from FE and 3 from BA.

Useful Tips (for placements and overall) :

I understand that the overall vibe of this resource might be a bit gloomy as I have probably ironed out a million pieces of **"things you need to be prepared for"**, but I'll try to make up for that here.

- **DSA:** DSA is extremely important but I feel like the amount of proficiency needed there is oversold. To join 92%+ companies visiting college you don't need to conquer Graphs or complete the 57 video Striver DP Series (although doing so is going to be beneficial only, no troubles there). You don't need to chase the 400, 500, 700 question targets. Having completed a popular problem roadmap like Strivers' SDE or Neetcode 150 will take you pretty far too. (This doesn't mean DSA is unimportant and the few top companies do rely on Dsa and Cp problems so be mindful of that, dialogues like these are subject to market risks, read all scheme related documents carefully).
- **Try to balance out your prep:** The reason I'm stressing this is because I've seen a lot of people sideline other topics like CS Fundamentals and Projects(same applies) and later get caught off guard. Try to allocate your time to time wisely and try to revise.
- **Resume:** Little buffs are fine but don't lie and don't write things you can't explain. It was more than once that I felt my knowledge of my own resume stretch to its absolute limits.

- **Read Interview Experiences (Duh):** You'll get to know how important which topic is for which company. Some ask for only DSA, some will grill you on your resume for hours. It's good to know where you're headed.
- **ML:** If you're interested in ML, try your best to work on projects and learn as much as possible. But don't depend on college for a placement in ML. The number of opportunities is limited and it is good to have a backup. Keep trying off campus too !
- **Always stay calm:** In a few interviews of mine I felt the tide flow against me and lost hope. This resulted in me getting frustrated and ending up somewhat arguing with the interviewer to defend my points. Never do this, always stay calm, you're not helping yourself anyhow behaving this way.
- **Rejections:** They are a part of life and you'll face them from time to time. Don't get disheartened, there's plenty of fish in the sea. You will get placed, don't stress too much.

ATB For Your Placements !!! Feel free to reach out to me on linkedin